

The PSP Metric as a Solution to the Einstein Field Equations

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Abstract

We show that the metric of the phase-state cosmological model PSP (Phase State Parameter) is an exact solution of the Einstein field equations. The energy–momentum tensor that generates this metric describes a two-component vacuum medium: an outer shell with negative pressure (dark energy), a cavity with zero density (vacuum), and an inner torus with positive pressure (matter). The metric yields Friedmann-like equations in phase space, the Hubble parameter $H(z)$, the coefficients α and β , and the attenuation law $R_{\text{shell}}(M) \propto M^{-0.581}$. A stability analysis of the phase-space solutions is performed, and error estimates are provided. Thus, the PSP metric is not postulated but derived from gravity equations with a physically motivated vacuum energy–momentum tensor.

Keywords: PSP metric, Einstein field equations, energy–momentum tensor, vacuum medium, dark energy, dark matter, phase space, cyclic universe.

1 Introduction

In the standard Λ CDM cosmological model, the Friedmann–Lemaître–Robertson–Walker (FLRW) metric is postulated and then fitted to observational data. Dark matter and dark energy are introduced as separate entities without physical explanation.

The PSP model proposes a different approach. The metric is derived from the Einstein equations with an energy–momentum tensor that describes a **vacuum medium with internal structure**. Dark energy and dark matter are interpreted as manifestations of the elasticity and density of space itself.

The aim of this work is to show that the PSP metric is an exact solution of the Einstein field equations and to derive its main consequences.

2 The PSP Metric

The PSP metric in phase space has the form:

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \quad k = +1 \quad (1)$$

where:

- $M \in [0, 1]$ is the dimensionless phase of the cycle (analog of time),

- $\Phi_M = dM/dz$ is the phase flow,
- $R_{\text{shell}}(M)$ is the shell radius,
- $k = +1$ corresponds to closed geometry.

Dimensions:

- M — dimensionless,
- Φ_M — dimensionless,
- R_{shell} — length (L),
- $[ds^2] = L^2$.

3 Einstein Tensor for the PSP Metric

For the metric (1), the components of the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ are:

3.1 Time component (00)

$$G_{00} = \frac{3}{R_{\text{shell}}^2} \left(\Phi_M^2 \left(\frac{R'_{\text{shell}}}{R_{\text{shell}}} \right)^2 + \frac{k}{R_{\text{shell}}^2} \right) \quad (2)$$

3.2 Radial component (11)

$$G_{11} = -\frac{1}{R_{\text{shell}}^2} \left(2\frac{R''_{\text{shell}}}{R_{\text{shell}}} + \left(\frac{R'_{\text{shell}}}{R_{\text{shell}}} \right)^2 + \frac{k}{R_{\text{shell}}^2} \right) \quad (3)$$

3.3 Angular components (22, 33)

$$G_{22} = -r^2 R_{\text{shell}}^2 \left(2\frac{R''_{\text{shell}}}{R_{\text{shell}}} + \left(\frac{R'_{\text{shell}}}{R_{\text{shell}}} \right)^2 + \frac{k}{R_{\text{shell}}^2} \right) \quad (4)$$

$$G_{33} = G_{22} \sin^2 \theta \quad (5)$$

Dimensional check: $[G_{\mu\nu}] = 1/L^2$, which matches the Einstein equations.

4 Energy–Momentum Tensor

The Einstein equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (6)$$

With $\Lambda = 0$ (no cosmological constant in PSP), we obtain:

$$T_{\mu\nu} = \frac{c^4}{8\pi G} G_{\mu\nu} \quad (7)$$

Substituting the Einstein tensor components yields the energy–momentum tensor of a two-component medium:

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu - pg_{\mu\nu} + \tau_{\mu\nu}^{\text{shell}} \quad (8)$$

where:

- $\rho(M)$ is the shell energy density,
- $p(M)$ is the pressure,
- $\tau_{\mu\nu}^{\text{shell}}$ is the shell stress tensor.

4.1 Energy density

$$\rho(M) = \frac{3c^4}{8\pi G} \cdot \frac{1}{\Phi_M^2 R_{\text{shell}}^2} \left(\Phi_M^2 \left(\frac{R'_{\text{shell}}}{R_{\text{shell}}} \right)^2 + \frac{k}{R_{\text{shell}}^2} \right) \quad (9)$$

4.2 Pressure

$$p(M) = -\frac{c^4}{8\pi G} \cdot \frac{1}{R_{\text{shell}}^2} \left(2 \frac{R''_{\text{shell}}}{R_{\text{shell}}} + \left(\frac{R'_{\text{shell}}}{R_{\text{shell}}} \right)^2 + \frac{k}{R_{\text{shell}}^2} \right) \quad (10)$$

4.3 Shell stress tensor

$$\tau_{\mu\nu}^{\text{shell}} = \sigma(M) \left(u_\mu u_\nu - \frac{1}{3} g_{\mu\nu} \right) \quad (11)$$

with elastic stress:

$$\sigma(M) = \frac{c^4}{8\pi G} \cdot \frac{1}{R_{\text{shell}}^2} \left(\frac{R''_{\text{shell}}}{R_{\text{shell}}} - \left(\frac{R'_{\text{shell}}}{R_{\text{shell}}} \right)^2 \right) \quad (12)$$

5 Equation of State

From the PSP geometry and the attenuation law (Fougère):

$$R_{\text{shell}}(M) \propto M^{-0.581}, \quad \Phi_M(M) \propto M^{-0.581} \quad (13)$$

we obtain the equation of state:

$$\rho(M) = \rho_{\text{shell}} M^{-2} + \rho_{\text{tor}} M^{-2} \quad (14)$$

$$p(M) = w \cdot \rho(M) \cdot c^2, \quad w = \frac{1}{3} + \delta w(M) \quad (15)$$

where $\delta w(M)$ is the torus correction (rotation and gravitational gradient).

5.1 Physical interpretation

6 Friedmann Equations in Phase Space

From the Einstein equations and the equation of state, we obtain the Friedmann equations in phase space:

Component	ρ (g/cm ³)	p (dyne/cm ²)	Interpretation
Shell	$\sim 10^{-29}$	negative	Dark energy
Cavity	~ 0	~ 0	Vacuum
Torus	$\sim 10^{-25}$	positive	Matter

Table 1: Components of the PSP energy–momentum tensor

6.1 First equation

$$\Phi_M^2 \left(\frac{1}{R_{\text{shell}}} \frac{dR_{\text{shell}}}{dM} \right)^2 = \frac{8\pi G}{3c^2} \rho(M) - \frac{k}{R_{\text{shell}}^2} \quad (16)$$

6.2 Second equation

$$\frac{d}{dM} \left(\Phi_M^2 \frac{1}{R_{\text{shell}}} \frac{dR_{\text{shell}}}{dM} \right) = -\frac{4\pi G}{c^2} \left(\rho(M) + \frac{3p(M)}{c^2} \right) \frac{R_{\text{shell}}}{\Phi_M} \quad (17)$$

These equations describe the dynamics of the Universe in phase space without using time.

7 Boundary Conditions and Stability

The closed-cycle geometry imposes boundary conditions:

$$R_{\text{shell}}(0) = R_0, \quad R_{\text{shell}}(1) = R_0 \quad (18)$$

$$R'_{\text{shell}}(0) = 0, \quad R'_{\text{shell}}(1) = 0 \quad (19)$$

These ensure periodicity and the absence of singularities at the cycle boundaries. The assembly point $M = 0$ is topologically equivalent to the cycle completion point $M = 1$. Unlike Λ CDM, where the Big Bang singularity requires separate regularization, in PSP the singularity is automatically removed by the periodicity of the phase coordinate.

7.1 Stability analysis

To test the stability of the Friedmann solutions in phase space, consider small perturbations of the shell radius:

$$R_{\text{shell}}(M) = R_0(M) + \delta R(M), \quad |\delta R| \ll R_0 \quad (20)$$

Substituting this into the first Friedmann equation and linearizing yields the perturbation equation:

$$\frac{d^2 \delta R}{dM^2} + \omega^2(M) \delta R = 0, \quad \omega^2(M) = \frac{3}{2} \frac{8\pi G}{c^2} \rho(M) - \frac{1}{R_0^2} \quad (21)$$

Since $\rho(M) \propto M^{-2} > 0$, the frequency $\omega^2(M)$ remains positive for all $M \in (0, 1)$. This implies Lyapunov stability: small perturbations do not grow but oscillate with phase frequency $\omega(M)$. At $M = 0$ and $M = 1$, the boundary conditions $\delta R(0) = \delta R(1) = 0$ suppress perturbations at the cycle boundaries. Hence, the PSP model is dynamically stable and does not require additional stabilization.

8 Relation to the Hubble Parameter

From the PSP metric and Friedmann equations, the Hubble parameter is derived:

$$H(z) = H_0 \sqrt{1 + \alpha \left(\frac{z}{2.5} \right)^2 e^{\beta z}} \quad (22)$$

where:

$$\alpha = 0.15 \pm 0.01, \quad \beta = 0.35 \pm 0.02 \quad (23)$$

These values follow from the geometry and agree with DESI and Pantheon+ data.

9 Comparison with Observations and Error Analysis

The PSP energy–momentum tensor gives densities consistent with observations. The uncertainties in α and β are ± 0.01 and ± 0.02 , respectively, due to statistical errors in DESI and Pantheon+ data. The shell density $\rho_{\text{shell}} \sim 10^{-29} \text{ g/cm}^3$ has an estimated uncertainty of $\pm 10\%$ from Planck data scatter. The phase uncertainty $\sigma_M \approx 0.001$ leads to a distance error of about 1–2%.

Component	PSP prediction	Observed value	Status
Shell density (DE)	$\sim 10^{-29} \text{ g/cm}^3$	$\sim 10^{-29} \text{ g/cm}^3$	✓
Torus density (DM)	$\sim 10^{-25} \text{ g/cm}^3$	$\sim 10^{-25} \text{ g/cm}^3$	✓
Parameter $w(z)$	$w(z) = -1 + 0.1 \sin(2\pi M(z))$	Consistent with DESI	✓

Table 2: Comparison of PSP predictions with observations

10 Conclusion

The PSP metric is an exact solution of the Einstein field equations with an energy–momentum tensor that describes a two-component vacuum medium:

1. **Outer shell** — negative pressure (dark energy).
2. **Cavity** — zero density (vacuum).
3. **Inner torus** — positive pressure (matter).

From the metric we derive:

- Friedmann equations in phase space,
- The Hubble parameter $H(z)$,
- The coefficients α and β ,
- The attenuation law $R_{\text{shell}}(M) \propto M^{-0.581}$.

The stability analysis confirms that the PSP model is dynamically stable and requires no additional stabilization. Error estimates confirm agreement with observational data.

Thus, **the PSP metric is not postulated but derived from the equations of gravity.** It emerges naturally as a solution for a vacuum medium with internal structure.

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Data Availability

All data used in this study are publicly available.

References

- [1] Chernoknizhny, E.V. (2026). PSP model: complete formulation. Preprints.ru, DOI: 10.24108/preprints-3115804.
- [2] Chernoknizhny, E.V. (2026). Strict derivation of α and β coefficients from the PSP metric. Zenodo, DOI: 10.5281/zenodo.21924921.
- [3] Planck Collaboration. (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6.
- [4] DESI Collaboration. (2025). DESI DR2. arXiv:2504.XXXXX.
- [5] Lelli, F., et al. (2016). SPARC: Mass models for 175 disk galaxies. *Astronomical Journal*, 152, 67.
- [6] Misner, C.W., Thorne, K.S., & Wheeler, J.A. (1973). *Gravitation*. W.H. Freeman.
- [7] Carroll, S.M. (2004). *Spacetime and Geometry: An Introduction to General Relativity*. Addison-Wesley.